

Zhi Yu

Researcher, Medical Image Analysis
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Research Profile

Researcher with training in bioengineering and advanced biomedical imaging, and one year of research experience in medical image analysis and clinical AI. Experienced in medical image segmentation, explainable image classification, image annotation, vision-language model adaptation, and temporal prediction. Currently focusing on clinically meaningful machine learning for biomedical imaging.

Education

University College London <i>MSc Advanced Biomedical Imaging - Merit</i>	Sep 2024 - Sep 2025 London, UK
Harbin Institute of Technology <i>BE Bioengineering - GPA: 3.6/4.0</i>	Sep 2020 - Jun 2024 Harbin, China

Publication

Surface Deformation Tracking in Monocular Laparoscopic Video <i>Medical Image Analysis</i>	2023 Z. Liu, W. Gao, J. Zhu, Z. Yu , Y. Fu.
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Research Experience

University of Catania <i>Research Intern in Medical Image Analysis</i>	Oct 2025 - Oct 2026 Catania, Italy
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Supervisor: Prof. Concetto Spampinato

Multimodal Model for Surgical Workflow Understanding

- Developed a multimodal framework combining laparoscopic images, temporal context, and structured surgical knowledge for current-state recognition and workflow anticipation.
- Built a hierarchical benchmark from cholecystectomy videos and adapted multiple vision-language models for current-state and future-state prediction.
- Developed an ensemble framework that improved next-state prediction on held-out surgical videos.

Explainable Paediatric Wrist-Fracture Assessment

- Developed a fracture-classification pipeline using the GrazPedWri-Dx paediatric wrist-radiograph dataset, including preprocessing, augmentation, and class-imbalance handling.
- Combined CNN-based classification with Grad-CAM, Integrated Gradients, and MedCLIP prompt matching to provide spatial and semantic explanations for model predictions.

University College London <i>MSc Project</i>	Apr - Aug 2025 London, UK
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Supervisor: Prof. Daniel Stuckey

Cine MRI Segmentation for Cardiac Function Assessment

- Analysed short-axis cine cardiac MRI data for automated segmentation of the left-ventricular blood pool and myocardium.
- Produced manual reference annotations in Horos and developed a UNet3+-based segmentation pipeline using TensorFlow.
- Derived and evaluated quantitative cardiac parameters using correlation analysis and Bland-Altman plots.

Harbin Institute of Technology <i>Graduate Design</i>	Sep 2023 - Jun 2024 Harbin, China
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Supervisor: Assoc. Prof. Wenpeng Gao

Laparoscopic Surgical Instrument Segmentation

- Developed a deep learning network combining U-Net architecture with Transformer for surgical instrument segmentation.
- Trained and evaluated on the EndoVis 2017 dataset using PyTorch, and outperformed several comparison networks.

Technical Expertise

Frameworks & Programming: PyTorch, TensorFlow, OpenCV, Python, C++
Medical Imaging: Horos, 3D Slicer

Additional Information

Languages: English (C1), French (B1), German (A1)